Paper 8 - Lattice Closure Template

CQE-paper-08
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Use high-dimensional lattice analogs as closure templates for transport without claiming global universality before local derivation.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-08.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-08\paper_kernel_manifest.json
- body: production\papers\CQE-paper-08\PAPER-BODY.md
- source: production\papers\CQE-paper-08\SOURCE.md
- formal: production\papers\CQE-paper-08\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-08\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-08\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-08.md

Paper 8 - Lattice Closure Template

Abstract

Paper 8 closes the first eight-paper window by proving the local lattice closure template used by the CQECMPLX suite. Papers 1-7 build local carrier, correction, registration, repair, path, causal, and bridge machinery. Paper 8 places that machinery inside a verified code/lattice ladder:

```
1 -> 3 -> 7 -> 8 -> 24
```

with powered terminal:

```
1^2 = 1
3^2 = 9
7^2 = 49
8 x 9 = 72
```

The theorem proves the local combinatorial scaffold. It does not promote the rootless Leech landing, Gamma72 polarization action, or uniqueness of all possible closure chains as closed theorems.

Claims

Claim 8.1. The chain `(1,3,7,8,24)` provides a verified local closure template for transporting local states into larger code/lattice frames.

Claim 8.2. The dimension-7 rung passes the Fano/Hamming identity checks.

Claim 8.3. The dimension-8 rung passes the extended Hamming/E8 seed checks.

Claim 8.4. The dimension-24 rung supplies Golay ingredients and `24 = 3 x 8` carrier geometry without proving the rootless Leech landing.

Claim 8.5. The powered terminal records `72 = 8 x 9` with sheet bound `K = 9`.

Claim 8.6. Gamma72 three-sheet transport is verified as transport, not as a closed Gamma72 landing proof.

Definitions

A **lattice closure template** is a sequence of finite code or lattice objects that lets a local state carrier be lifted into a larger transport frame while preserving the proof boundary of every step.

The Paper 8 template is:

```
D1 raw bit                 -> 1
S3 / three-cell carrier    -> 3
Fano / Hamming / octonion  -> 7
extended Hamming / E8 seed -> 8
Golay / Leech ingredient   -> 24
Nebe sheet bound           -> 72
```

A **local certified fact** is a claim checked at a fixed dimension by a finite verifier.

A **global landing claim** is a stronger statement that a local certified fact has been glued into a terminal lattice object such as the rootless Leech lattice or a Gamma72 action.

Theorem 8.1 - Local Lattice Closure Template

The finite code chain `(1,3,7,8,24)` and powered terminal `72 = 8 x 9` provide a verified local closure template for CQECMPLX transport. Every admitted rung is backed by a finite combinatorial check, and every unproved global landing is preserved as an explicit proof boundary.

Proof

The verifier establishes the following local facts.

At dimension 7, the `(7,4,3)` Hamming code has 16 codewords, minimum weight 3, weight distribution:

```
{0:1, 3:7, 4:7, 7:1}
```

and exactly seven weight-3 codewords whose supports are the seven Fano-plane lines. This identifies the Hamming, Fano, and octonion-imaginary incidence structure at the local dimension-7 rung.

At dimension 8, the `(8,4,4)` extended Hamming code has 16 codewords, minimum weight 4, weight distribution:

```
{0:1, 4:14, 8:1}
```

all weights divisible by 4, and pairwise self-orthogonality. This supplies the self-dual doubly-even binary code used as the E8 Construction A seed.

At dimension 24, the Golay generator layer has 12 length-24 rows, zero generator-pair orthogonality errors, and the carrier geometry:

```
24 = 3 x 8
```

This supplies Golay ingredients and three-copy D4 carrier geometry. It does not by itself prove a rootless Leech landing.

For the powered terminal:

```
1^2 = 1
3^2 = 9
7^2 = 49
8 x 9 = 72
```

The verifier records `sheet_K_bound = 9` and dimension-72 extremal minimum norm:

```
2 * floor(72/24) + 2 = 8
```

The Gamma72 contract checks 260 payloads and confirms exact round trips into three Leech sheets, while also recording:

```
polarization_matrices_supplied = false
gamma72_landing_proved = false
```

Together these checks prove the local closure template and its audit boundary.

Receipt

The primary executable receipt is:

```
production/formal-papers/CQE-paper-08/verify_lattice_closure_template.py
production/formal-papers/CQE-paper-08/lattice_closure_template_receipt.json
```

The receipt status is `pass`. It verifies:

```
chain_passes                                        = true
fano_hamming_identity_passes                        = true
extended_hamming_e8_seed_passes                     = true
golay_24_ingredient_passes_without_leech_overclaim = true
powered_72_sheet_bound_passes                       = true
gamma72_transport_passes_without_landing_overclaim = true
leech_landing_overclaim_rejected                    = true
gamma72_landing_overclaim_rejected                  = true
```

Falsifiers

The verifier rejects:

```
Golay ingredients alone prove the rootless Leech landing
three Leech-sheet round trips prove the Gamma72 polarization action
the chain proves that no other higher-dimensional closure template exists
```

These statements exceed the certified local checks.

Role in the Suite

Paper 8 receives the first block:

```
LCR carrier
-> correction surface
-> D4/J3 coordinate surface
-> boundary repair
-> path carrier
-> causal code
-> discrete-continuous bridge
-> lattice closure template
```

It also wraps the block back to Paper 1 by showing how the local carrier can enter a higher-dimensional scaffold without changing the proof status of any unclosed global landing.

Open Audit Boundaries

- A rootless Leech landing proof requires an explicit glue-action verifier.
- A Gamma72 landing proof requires selected Hermitian polarization matrices.
- A cold-start map from arbitrary depth to a Niemeier/Leech fingerprint

remains outside this paper.

- The claim that this is the only possible closure chain is not part of the

theorem.

Conclusion

Paper 8 proves the first-block closure scaffold. Local lattice facts are certified, their transport role is explicit, and stronger global landings are kept visible as named audit boundaries rather than hidden inside the theorem.